

Farahnaz Amiri

Curriculum Vitae | Specialized in Number Theory & Arithmetic Statistics

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Research Interests

- **Primary:** Arithmetic Statistics, Class Group Structures, Higher Composition Laws.
- **Secondary:** Arithmetic Geometry, Rank of Elliptic Curves.

Academic Positions

Researcher (Visiting Scholar) — IPM, Tehran, Iran **Sep 2025 – Present**
Focusing on Arithmetic Geometry.

Postdoctoral Fellow — BIMSA (Supported by Tsinghua University) **Jul 2023 – Sep 2025**
Research in Number Theory; Host: Prof. Yau.

Education

Ph.D. in Pure Mathematics — Tarbiat Modares University **2016 – 2023**
Thesis: *A new method to determine totally imaginary Galois fields with class number one of a given degree.* |
Advisor: Prof. Ali Rajaei

M.Sc. in Pure Mathematics — Sharif University of Technology **2013 – 2015**
Thesis: *Average Rank of Elliptic Curves.*

B.Sc. in Pure Mathematics — University of Mashhad **2009 – 2013**

Publications & Preprints

Journal Publications

1. **The Class Number One Problem for Imaginary Octic Non-CM Extensions of $\mathbb{Q}$** , *Bulletin of the Iranian Mathematical Society (BIMS)* 49 (2023), no. 4, 56–65. (with A. Rajaei)

Preprints

2. **Computing p -Class Group Structure in Real Quadratic Fields: A New Approach**, *Submitted (2026)*.
Available at arXiv: [2601.19288 \[math.NT\]](https://arxiv.org/abs/2601.19288)

Works in Preparation

3. **Higher Composition Laws for Non-Galois Number Fields**, *In preparation*.
4. **Higher Composition Laws and the 5-Part of the Class Group of Real Quadratic Fields**, *In preparation*.

Invited Talks & Presentations

Selected Invited Seminar Talks

- *Gauss & Bhargava Higher Composition Laws: A New Perspective*, **IPM Workshop on Elliptic Curves**, Jolfa, Iran, Nov 2025.
- *On Composition Laws*, **Tsinghua BIMSA Number Theory Lunch Seminar**, Beijing, China, May 2025.
- *Composition Laws: A New Perspective*, **YMSC Workshop, Tsinghua University**, Beijing, China, Apr 2025.
- *Extending the Resolution Process for Gauss's Conjectures*, **BIMSA Workshop**, Beijing, China, Dec 2024.
- *The Average Number of Class Numbers of Quadratic Fields*, **Tsinghua BIMSA Number Theory Lunch Seminar**, May 2024.
- *Biquadratic Forms and Class Number Problems*, **Weekly Seminar, Tarbiat Modares University**, Tehran, Jan 2024.
- *Imaginary Fields with Class Number One*, **Tsinghua BIMSA Number Theory Lunch Seminar**, Oct 2023.
- *New Families of Imaginary Fields with Class Number One*, **Postdoc Workshops, YMSC**, Oct 2023.

Conference Presentations

- *Non-Unique Factorization of Dihedral Extensions of Degree 32*, **4th International Conference on Mathematics and Basic Sciences**, Aug 2021.
- *Class Number One: From Gauss to the Present*, **Sharif Women in Math, Sharif University of Technology**, Jan 2021.

Grants, Awards & Honors

Project Grant, Beijing Natural Science Foundation (BJNSFC) Grant No. LS24064: Principal Investigator. Funding: 100,000 CNY/year.	2024 – 2026
Travel Grant, Natural Science Foundation (NSF) To attend Analysis and Number Theory Conference, Pittsburgh, USA.	2024
Performance Research Bonus, BIMSA Awarded for excellence in postdoctoral research.	2024 – 2025
Langlands Program Scholarship, IPM Competitive scholarship for advanced research.	2015 – 2016
Ninth Rank, National Mathematical Olympiad Among all undergraduate math students in Iran.	2013

References

- **Prof. Mehrdad Shahshahani (IPM)** - shahshah@ipm.ir

- **Prof. Ramin Takloo-Bighash** (UIC) - rtakloo@math.uic.edu
- **Prof. Artan Sheshmani** (BIMSA & Harvard) - artan@bimsa.cn
- **Prof. Ali Rajaei** (Tarbiat Modares) - alirajaei@modares.ac.ir